THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The side of a rhombus is 35 cm and one diagonal is 42 cm. The other diagonal (in cm) is: 1

(a) 58 **(b)** 57 **(c)** 56 **(d)** 28

2. Two vertical poles of heights 45 m and 24 m stand 28 m apart on level ground. The distance between their tops (in m) is: 1

(a) 49 **(b)** 36 **(c)** 33 **(d)** 35

3. The area of a right triangle with hypotenuse 35 cm and one leg 21 cm (in cm^2) is: 1

(a) 315 **(b)** 588 **(c)** 294 **(d)** 367

4. Which set of numbers is a Pythagorean triplet? 1

(a) 21, 74, 75 **(b)** 21, 72, 76 **(c)** 22, 72, 75 **(d)** 21, 72, 75

5. An isosceles triangle has equal sides 50 cm and base 60 cm. Its height (in cm) is: 1

(a) 41 **(b)** 39 **(c)** 20 **(d)** 40

6. A rectangle has a diagonal of 50 cm and one side 30 cm. Its perimeter (in cm) is: 1

(a) 160 **(b)** 140 **(c)** 142 **(d)** 70

7. The side of a rhombus is 50 cm and one diagonal is 60 cm. The other diagonal (in cm) is: 1

(a) 80 **(b)** 40 **(c)** 82 **(d)** 81

8. Two vertical poles of heights 50 m and 20 m stand 40 m apart on level ground. The distance between their tops (in m) is: 1

(a) 48 **(b)** 50 **(c)** 51 **(d)** 70

9. The area of a right triangle with hypotenuse 50 cm and one leg 30 cm (in cm^2) is: 1

(a) 750 **(b)** 630 **(c)** 600 **(d)** 1200

10. Which set of numbers is a Pythagorean triplet? 1

(a) 40, 52, 65 **(b)** 39, 52, 66 **(c)** 39, 54, 65 **(d)** 39, 52, 65

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The side of a rhombus is 45 cm and one diagonal is 54 cm. Find the other diagonal. 2

12. An isosceles triangle has equal sides of 35 cm and base 42 cm. Find its height. 2

13. The diagonal of a rectangle is 50 cm and one side is 14 cm. Find the other side and the area. 2

14. The side of a rhombus is 58 cm and one diagonal is 80 cm. Find the other diagonal. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A ladder 26 m long reaches a point 24 m up a wall. The foot is then pulled away so that it is 24 m from the wall. How far down does the top of the ladder slip? 3

16. Show that the triangle with sides 30 cm, 40 cm and 50 cm is right-angled and find its area. 3

17. The altitude drawn to the hypotenuse of a right triangle with legs 30 cm and 40 cm: find the hypotenuse and the length of this altitude. 3

OR

A ladder 39 m long reaches a point 36 m up a wall. The foot is then pulled away so that it is 36 m from the wall. How far down does the top of the ladder slip? 3

18. Show that the triangle with sides 14 cm, 48 cm and 50 cm is right-angled and find its area. 3

OR

The altitude drawn to the hypotenuse of a right triangle with legs 14 cm and 48 cm: find the hypotenuse and the length of this altitude. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A man walks 30 m east and then 40 m north. Find 5
(i) the straight-line distance of his final position from the start,
(ii) the distance saved by going straight,
(iii) the total distance if he returns to the start by the straight path.

OR

An isosceles triangle has equal sides of 45 cm and base 54 cm. Find its height, area and perimeter. 5

20. A man walks 14 m east and then 48 m north. Find 5
(i) the straight-line distance of his final position from the start,
(ii) the distance saved by going straight,
(iii) the total distance if he returns to the start by the straight path.

OR

An isosceles triangle has equal sides of 58 cm and base 80 cm. Find its height, area and perimeter. 5

END OF QUESTION PAPER — Total: 40 Marks

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